

Are you Seeing what I'm Seeing?: Perceptual Issues with Digital Twins in Virtual Reality

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ABSTRACT

With the availability of data and visualization capabilities, digital twins have grown in use cases and entered the virtual reality (VR) realm. Utilizing VR enhances the value of digital twins by enabling training, testing, and refinement of a system, through support of a user's workflow when interacting with the system. However, VR is not fully representative of real-world environments, particularly in areas such as visual perception, proprioception, vestibular response, and perceptions of information. The simulation and training domain has learned which ways VR can benefit users and which ways it falls short—allowing a rich domain of information for the digital twin world to leverage. While many of these differences can be mitigated through targeted environmental design or hardware selection, these differences in perceptual capabilities and feedback negatively impact the validity of VR digital twins. These impacts could range from user discomfort, injury, negative transfer to the real world, and poor design decisions based on unrepresentative digital twins. This paper will discuss the current use cases of digital twins in VR that center on human interaction, common perceptual concerns in VR uncovered in simulation research, and how these issues could impact a digital twin's representativeness. As the applicability of digital twins in VR expands, mitigations for perceptual issues in VR and open areas for improvement in VR for digital twins are presented.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality Computing methodologies—Computer graphics—Graphics systems and interfaces—Perception

1 HUMAN INTERACTION DIGITAL TWINS USE-CASES

Digital twins are commonly referred to as mirrors or replicas of a real-world object or system built upon large amounts of data that also update based on real-world performance [1, 2]. Advancements in computing and visualization software have allowed for immersive and realistic simulations of real-world systems. The first and primary use of digital twins has been in the manufacturing domain enabling remote monitoring and control, low-cost safety testing and evaluation, predicted maintenance and support, and system training and documentation [1, 3]. Due to its ability to iterate and update itself (based on predicted manufacturing performance and actual) the manufacturing process can be analyzed and improved. Some additional use cases are presented in Table 1. These use cases present an opportunity to leverage complex models of digital twins as well as visualization tools such as VR to improve real world systems. However, as digital twins become commonplace in many of these system-dependent domains, human perceptual and cognitive issues should be considered to ensure usability and safety.

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2 PERCEPTUAL ISSUES IN VIRTUAL REALITY

VR puts users into a state of altered perception, particularly in domains of visual and proprioceptive processing. Human perception and interaction within VR can be defined as a multi-layered fidelity framework, where the layers of interest to this work are the sensory and perceptuomotor layers [12]. The sensory layer defines the understanding of information received through human visual, tactile, proprioceptive, and olfactory systems. The perceptuomotor layer focuses on the mapping of real-world motor actions and their translation within the virtual environment. One key challenge in this space is the limitations associated with the hardware itself. Even if the digital twin can model the visual, tactile, and proprioceptive cues, the hardware responsible for presenting it to the user must be able to match the level of fidelity similar to the real world [8]. The following sections will outline current perceptual and cognitive issues associated with VR and how they could impact the usability effectiveness of digital twins in novel use cases.

2.1 Changes in Field of View

A well-documented concern is that a reduction in a user's field of view (FOV), which can occur when using a VR head-mounted display (HMD), can negatively impact a user's ability to complete a task in a timely manner [13]. In construction domains, a reduced field of view in a planning phase may result in greater head movements necessary to take in all necessary 3d renderings. Additionally, the reduced FOV may result in missing critical information during planning and decision making. In regard to the effect of a reduced FOV on a user's behavior and decision-making capabilities, it is necessary to understand the usage of peripheral vision towards underlying motor tasks. Specifically, balance and the mechanics of visual search.

Balance is a combination of input from the visual and vestibular systems. As individuals age, they rely more on input from their visual system to balance, as compared to signals from their vestibular system [14]. Similarly, studies suggest that when in VR, input from the user's visual system may become less reliable than their vestibular system [15]. Corollary, studies on peripheral vision suggest that users rely heavily on their peripheral vision to maintain balance during standing tasks [14, 16]. As such, while under investigation, there is a likelihood that a reduced field of view in VR can reduce the stability of users. Consequently, actions in VR reliant on maintaining and assessing balance are potentially unreliable.

The functional severity of balance changes within VR are unknown. Generally, studies on balance in VR rely on standing tests, which suggest that an individual's balance in VR is not worse than their balance when their eyes are closed [15]. However, it is understudied whether the severity of balance alterations in VR are consistent when the individual is in motion.

Balance alterations are of significant concern for work related to balance or simulating user motion [17]. This is particularly problematic when using VR to train and assess individuals to use systems that rely on postural control and stability, such as recent investigations into the use of VR to train surfing skills [18], as users could be training under non-representative stability conditions.

A second consequence of a reduced FOV, is a potential impairment of spatial localization, navigation, and visual search capabil-

Table 1: Use Cases of Digital Twins involving Human Interaction.

Domain	Purposes
Manufacturing	Conceptualize complex systems, compare against requirements, and collaborate [1, 4].
Construction	Plan and verification of progress or use as a collaborative tool [5].
City Planning	Identify areas of unsatisfactory city characteristics and plan for infrastructure [6].
Maintenance	Visualization tool of information both during task and in distributed locations [7].
Medical	View and track patient vitals and health for predictive analysis and treatment [8, 9].
Training	Synchronize with live systems to practice procedures in a safe environment [10, 11].

ities. Studies show that when performing active navigation tasks, individuals rely on their peripheral vision to map and update novel paths [19, 20]. Without access to their peripheral vision field, the pathways individuals take in new environments can become suboptimal due to the loss of environmental awareness.

Studies on navigation impairment often focus on severe reductions in FOV (e.g., 10° FOV), and even then, FOV-induced changes in an individual's pathing can require the layout of an environment to be complex or unpredictable [21]. As such, while this may cause an impact on routing and navigation within VR, it is unlikely to be a severe or noticeable difference if one does exist. This is not the case for the impact of FOV on visual search capabilities, which are better studied in virtual and augmented reality domains. A 2019 study on the effect of FOV reduction for the execution of a visual search task found that a 35° FOV (which matches the specifications of a Microsoft HoloLens) did alter visual search strategies, and reduced participants' task completion time by up to 37% [22].

Due to the cost and difficulty of procuring large-scale environments to test navigation strategies in the real-world, VR is considered an option to simulate representative environments for navigation [23]. However, the majority of commercial VR HMDs reduce the user's FOV, and due to the changes that a reduced FOV may cause to visual search and navigation strategies, it cannot be guaranteed that work in this domain is valid.

2.2 Vergence Accommodation Conflict

When the human eye focuses on visual information, it performs two processes in parallel, (1) accommodation, during which the shape of the lens in the eye will change to optimize the amount of light coming from the targeted visual information based on distance, and (2) vergence, in which the eye will rotate to keep the object centered in the fovea of the pupil. Because vergence and accommodation generally happen in parallel, they are jointly referred to as the vergence-accommodation reflex. A diagram of this process is shown in Figure 1.

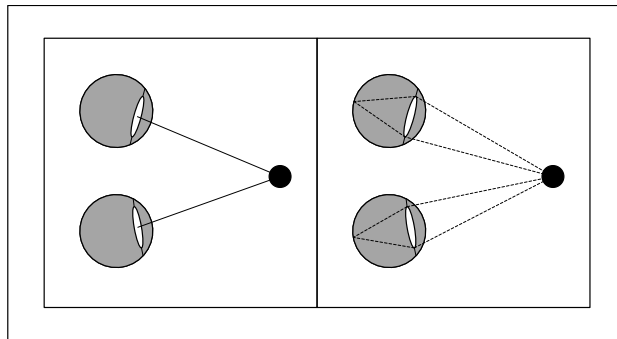


Figure 1: Vergence (left) the pupils rotated to keep an object centered on their retina; Accommodation (right) the lens in the eye changes shape to optimize the light coming from the target object. Traditionally vergence and accommodation only occur in tandem.

Stereoscopic HMDs project visual information from a virtual

environment to a stationary lens mounted in front of the wearer's eyes. However, this method of simulating depth interferes with the vergence-accommodation reflex. While vergence will still occur, there is no need to adjust the accommodative distance of the lens, because all virtual objects are located the same distance from the eye. Figure 2 highlights how, while vergence will still cause the eyes to rotate to the perceived location of an object, accommodation never changes from the lens of the HMD.

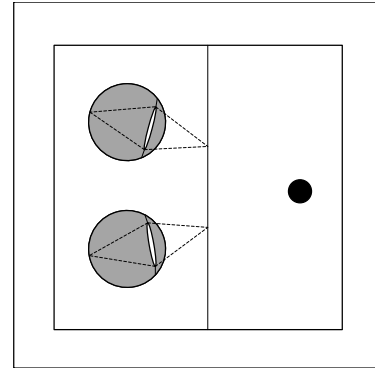


Figure 2: Shows the conditions of a VR HMD when an object is projected to a static lens. While vergence occurs, and rotates the eyes to where it believes the object is, the accommodative focal distance will always be the lens.

This forcible disconnect of the vergence-accommodation reflex is referred to as vergence-accommodation conflict (VAC) [24]. VAC is a stressful state to maintain for the ocular system, and over time can result in visual fatigue (e.g., eyestrain, headaches, etc.) if it is not mitigated. This build-up of visual fatigue over time can affect a user's cognitive state and impair their decision-making. This could be of particular concern for construction and maintenance domains that aim to utilize augmented reality digital twins to assist workers on the job.

The effects of visual fatigue and the other symptoms of VAC are indirect, but severe. A 2019 study investigated the effects of VR HMDs on decision-making and cognitive processing [25]. This evaluation used a multi-choice reaction time test, where users had to visually monitor information, then select the appropriate response. During this study it was noted that users who had executed the VR task before the assessment demonstrated a marked increase in the time required to make decisions. While the underlying mechanism requires further analysis, the authors hypothesized that this increase in processing time may be a result of recovery from phoria adaptation. Regardless, this study does suggest that extended use of VR HMDs can negatively impact decision-making capabilities, potentially reducing the validity of VR environments for behavioral testing over time.

2.3 Ocular-vestibular mismatch

An issue associated with the onset of cyber-sickness is the mismatch between a user's ocular and vestibular system. Most commonly, this

will occur when the optical flow projected by a VR system suggests that the user will be in motion, and the user's inner ear does not demonstrate an analogous motion response. This can be as explicit as experiencing a motion in VR while stationary, or as subtle as viewing an imperceptible sinusoidal motion pattern (i.e., visual perturbances that can occur due to tracking failures or irregularities) [26, 27]. Current hardware constraints can exacerbate this such as limited interpupillary distance ranges within VR headsets, which can distort viewing angles per eye, and has been cited as a primary cause of increased cyber sickness ratings amongst women [28]. There may be additional perceptual issues as a result.

2.4 Loss of environmental cues

When a VR system does not accurately replicate visual cues used to assist with localization, it can impact a user's ability to understand spatial relationships in the virtual environment [29]. A key example of this is the misprojection or a lack of shadows in a virtual environment. In the case of safety training systems, machinery may be represented and synced with machinery in the real world offering a realistic training scenario [10]. However, lack of shadows or lighting in these environments could lead to inaccurate mental models regarding the real-world machinery's operating space—potentially leading to accidents and injury.

As a result of spatial misestimation, among other issues (e.g., narrow FOV, altered self-perception of size), studies show that users wearing a VR HMD are unable to properly estimate the distance or size of objects. This makes VR with an HMD particularly ill-suited for testing decision-making related to spatial relationships between objects. Corroborative driving studies show differences in driving behavior between real-world and representative VR HMD-driven virtual environments [30]. Spatial misestimation may then also lead to inaccurate training or accidents.

3 MITIGATION STRATEGIES AND OPEN ISSUES

Many of these issues can be avoided through intelligent system design and hardware selection, but some represent open research questions that require additional research to mitigate.

3.1 Limited field of view

To mitigate the issues caused by a narrow FOV, the VR environment should closely resemble the user's FOV range [16]. However, this is not always the best solution. A selection of contemporary commercial VR HMDs feature a wider FOV (e.g., PIMAX, StarVR), and can simulate peripheral vision more effectively, but studies suggest that they are unable to fully resolve peripheral-adjacent visual distortions, and could increase the likelihood of cybersickness [31, 32]. It is not yet fully known how these distortion effects visual recognition, and whether it will have secondary impacts on balance, visual search, and navigation strategies. CAVEs and platform-specific simulator (e.g., flight simulators) are unlikely to suffer from the type of distortion.

3.2 Vergence-Accommodation Conflict

VAC occurs when stereoscopic displays simulate depth. While it is most apparent when using VR HMDs, many CAVEs will also leverage stereo display to enhance feelings of depth. There is ongoing research into commercially affordable display options that reduce the elicitation of VAC [33], but the only mitigation currently commercially available is intelligent system design.

VAC occurs because the perceived accommodation of a virtual object does not match the object's perceived vergence depth. As such, VAC symptoms are abated when vergence and accommodation depths align properly. Studies recommend that VAC symptoms can be reduced by providing eyes an opportunity to rest after prolonged elicitation of VAC conditions. This can be done by either offering users a break from using the system [34, 35], or by aligning virtual objects so that their vergence and accommodation depths match for

a period of time (i.e., have the user focus at the correct focal distance) [33, 36]. For stereo VR systems that simulate depth, the focal distance is often static and can vary across headsets, so the hardware specific accommodation depth should be understood beforehand.

There is a caveat, that prolonged usage of a stereoscopic VR system will induce phoria adaptation. Phoria references differences in alignment between the eyes and occurs when the eyes are not looking at the same object (i.e., vision is unfused). It is possible that breaking the vergence-accommodation reflex simulates the state of both eyes looking at different objects in space [37–39].

When phoria is induced by freezing the focal plane, which is what occurs within VR, after a variable amount of time, accommodation for the eyes will remain at the frozen focal plane independent of vergence. This process, which is a type of phoria adaptation, can reduce feelings of VAC symptomology [24]. Studies suggest that phoria adaptation will prolong the amount of time taken for the ocular system to recover after the stereoscopic VR experience has ended. The occurrence of phoria adaptation can be prevented (if desired) by taking frequent breaks, although this can exacerbate immediate symptoms [36]. Further hardware-based solutions to VAC are under active research [33].

3.3 Ocular-Vestibular Mismatch

Ocular-vestibular mismatching can be mitigated through both hardware and system design methods. Hardware mitigations include simulator platforms that replicate relative motion experienced by the user within the virtual environment [40], and body-worn systems to replicate missing optical flow and/or vestibular signatures [41, 42].

Design methods to mitigate Ocular-vestibular mismatching symptoms include locomotive mapping, that enables users to elicit figurative or similar vestibular responses to what they experience through their optical flow. Examples include redirected walking that manipulates the optical flow to redirect real-world walking by leveraging differences in processing between the ocular and vestibular systems [43].

3.4 Depth and size misestimation

Recent research on mitigation of size and depth misestimation errors in VR seek to resolve this problem by resizing the user's perspective of their own size within the world, which adjusts the user's perception of spatial relationships. Approaches of interest include providing users a hand as their control medium in the environment, and resizing this hand until spatial relationships match expectations [44, 45].

4 DISCUSSION

Although the distinction can be made that simulation is for training, as a static environment, and digital twins are for the development and evaluation of systems, these perceptual issues will impact both. Use cases such as NASA's Apollo program and the Iron Bird utilized digital twins—flown by humans—to determine any changes needed in system design [11]. The perceptual and cognitive issues interacting with digital twins should be considered when designing systems around human performance with digital twins.

VR systems can result in many perceptual issues for users. Resulting impacts to human design decisions based on VR representations of digital twins are of concern. FOV limitations can result in poor balance, altered navigation or search strategies, and missed information. Lack of environmental cues, such as shadows, can lead to inaccurate estimations of distance and size potentially leading to injury or poor estimations with the real-world system. Finally, mismatches in ocular and vestibular systems with the virtual world displays will cause fatigue, cyber sickness, and poor decision making. Each of these issues can alter the way a human interacts with digital twins which could impact real-world system designs based on digital twins.

If a user's understanding of a virtual environment differs from their understanding of the real-world scenario it is built to simulate, it cannot be guaranteed that the user will express identical behaviors and decisions between the two environments. Additionally, the digital twin's representation of the environment is only as good as the sensors that collect the original data [2]. Information that is not perfectly matched will lead to perceptual differences once the user takes any training or decision making into the real world. User perceptions of digital twin accuracy can degrade if the digital twin does not match the user's mental model which could lead to misuse or disuse of digital twin systems [5].

Digital twins in VR are emerging—but not necessarily optimized. Mitigation strategies for perceptual issues for VR include reducing the FOV, reducing required time in VR, utilizing altered navigation methods, and incorporating references uses to ensure accurate size and distance representation. Hardware and design of these digital twins—and the way humans interact with them—could help prevent perceptual issues and ultimately inaccurate design decisions. More research is needed to design hardware and software that supports extended use and accurate perception. Additionally, VR digital twin use cases still need interface design considerations to support sense-making of abstract data and decision support based on multiple sources of information [6, 7]. If information from these digital twin environments can provide it in a way that supports human perception and decision making, the benefits for digital twin technology can really advance.

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